



HUMAN BRAIN TUMOR CLASSIFICATION USING CNN

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**Submitted to
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1. Overview

The project focuses on the detection and classification of brain tumors from MRI images using deep learning techniques, specifically Convolutional Neural Networks (CNN). The goal is to automate tumor identification, reducing dependency on expert neurosurgeons and minimizing human error, especially in regions lacking skilled professionals.

2. Motivation

Brain tumors vary greatly in size, shape, and location, making manual analysis complex and time-consuming. An automated system can:

- Detect the presence of a tumor.
- Classify the tumor into one of the main types (glioma, meningioma, pituitary, or no tumor).

Such a system can significantly improve early diagnosis, reduce errors, and assist patients in making informed decisions quickly.

3. Problem Definition and Objectives

3.1 Definition

The project aims to detect and classify brain tumors using CNN as part of deep learning techniques and provide positional information for identified tumors.

3.2 Objectives

1. Classify MRI as tumor or no tumor.
2. Classify detected tumors into one of four categories: glioma, meningioma, pituitary, or no tumor.

4. Project Scope and Limitations

4.1 Scope

- Data preprocessing and cleaning.
- Building multiple CNN models and transfer learning (TL) approaches.
- Selecting the best model with optimized hyperparameters.
- Deployment on cloud platforms (AWS).
- Visualizing tumor location on MRI.
- Minimizing medical negligence.

4.2 Limitations

- Only jpg and png MRI formats are supported.
- Non-MRI images may not be correctly identified.
- Only tumor classification is considered; other medical conditions are not evaluated.

5. Methodology

5.1 Handling Underfitting

Description: Occurs when the model has not trained sufficiently or lacks complexity.

Solution: Increase model complexity, add more parameters, and train longer.

5.2 Handling Overfitting

Description: Model performs well on training data but poorly on unseen data.

Solution: Use more data, cross-validation, early stopping, regularization, and ensemble methods.

5.3 Infrastructure Considerations

Description: GPU limitations may cause long training times or crashes.

Solution: Use platforms like Google Colab or AWS Sagemaker with sufficient GPU memory (up to 50GB).

6. Experimental Results

6.1 Model Performance

- Training Accuracy: 96.36%
- Validation Accuracy: 74.87%
- Test Accuracy: 74.87%
- Test Loss: 6.1132

Observation: The model shows high training accuracy but a noticeable gap with validation accuracy, indicating some overfitting.

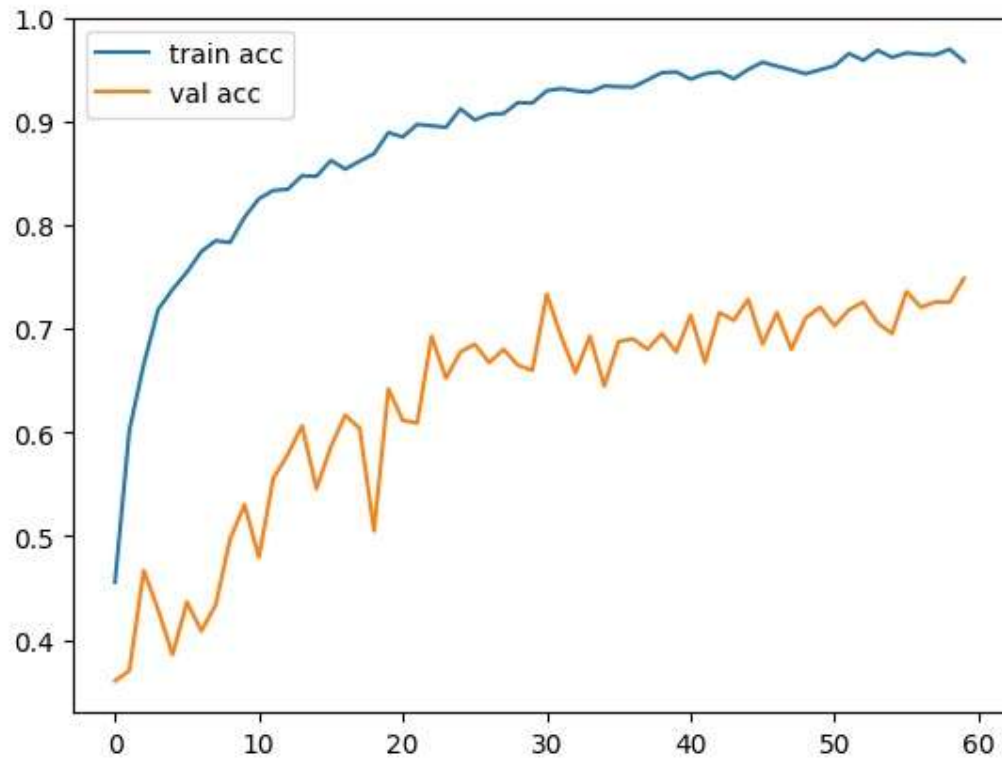


Figure 1 Training Accuracy

6.2 Confusion Matrix

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- Glioma tumor is mostly misclassified.
- Meningioma and pituitary tumor classes show strong classification accuracy.
- No tumor class is correctly classified in almost all cases.

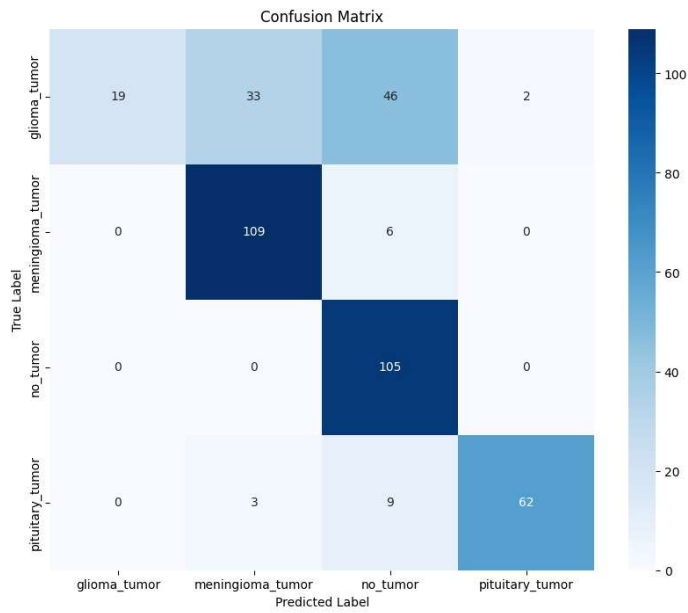


Figure 2 ConfusionMatrix

6.3 Classification Report

Class	Precision	Recall	F1-Score	Support
Glioma tumor	1.00	0.19	0.32	100
Meningioma tumor	0.75	0.95	0.84	115
No tumor	0.63	1.00	0.77	105
Pituitary tumor	0.97	0.84	0.90	74
Accuracy			0.75	394
Macro avg	0.84	0.74	0.71	394
Weighted avg	0.82	0.75	0.70	394

6.4 Accuracy Curve

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- Training accuracy steadily increases over epochs, while validation accuracy fluctuates and plateaus around 0.74.
- Indicates potential overfitting and need for regularization or data augmentation.

7. Recommendations and Future Work

1. Data Augmentation:

- Apply rotation, flipping, scaling, and brightness adjustments to increase dataset diversity.
- Helps the model generalize better and reduce overfitting.

2. Transfer Learning:

- Use pre-trained CNN architectures like VGG16, ResNet, or EfficientNet for feature extraction.
- Fine-tune the model for better performance on small datasets.

3. **Explainable AI (XAI):**

- Use techniques like Grad-CAM or LIME to highlight regions of the MRI influencing predictions.
- Improves transparency and trust in automated diagnosis.

4. **Ensemble Models:**

- Combine predictions from multiple models to improve classification accuracy and robustness.

5. **Hyperparameter Optimization:**

- Experiment with learning rate schedulers, batch sizes, and optimizers to improve validation performance.

6. **Deployment Enhancements:**

- Integrate a web interface or mobile app for real-time MRI analysis.
- Allow doctors to annotate predictions and feedback to continuously improve the model.

7. **Expand Dataset:**

- Include more images from multiple sources to reduce bias and increase model generalization.

8. Conclusion

- The CNN model successfully classifies meningioma, pituitary, and no tumor categories.
- Glioma classification requires improvement due to low recall.
- With recommended improvements like data augmentation, transfer learning, and explainable AI, the system can become a reliable assistant for brain tumor detection, reducing manual workload and improving early diagnosis.

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